

Research Progress Check Point 3

Reliability-Aware Rank Aggregation for Deployment-Oriented Predictive Maintenance Model Selection

1. Research Progress Overview

During Research Progress Check Point 3, the project progressed from the initial development and exploratory validation of Reliability-Aware Deployment Rank Aggregation (RDRA) established during Check Point 2 towards a more formally defined, statistically validated and publication-oriented research methodology.

Predictive maintenance increasingly uses data-driven and machine-learning methods to support fault prediction, machinery-health assessment and Remaining Useful Life (RUL) estimation [1]-[5]. However, selecting a model for practical deployment involves more than identifying the configuration with the highest predictive accuracy. Robustness to sensor degradation, reliability of predicted probabilities, computational requirements and asymmetric operational consequences can all affect whether a model is practically suitable [9], [11]. More broadly, algorithm selection and multi-criteria decision-making research demonstrates that alternatives can be evaluated differently depending on the criteria and aggregation method used [12]-[15].

Check Point 2 established the central research problem addressed by this project: deployment-oriented predictive-maintenance evaluation can produce conflicting criterion-specific model rankings, while the empirical evidence supporting these rankings may differ in reproducibility. Existing multi-criteria and rank-aggregation research provides mechanisms for combining alternative rankings [12], [13], [15], [18], but the present research investigates a more specific question: whether the empirical reproducibility of each deployment-criterion ranking can be used as reliability information when constructing a deployment-oriented consensus.

Consequently, RDRA is treated as the central methodological contribution. Equal Borda is retained as the principal aggregation baseline, while robustness analysis, criterion perturbation, weight sensitivity, data-quality responsiveness and downstream preference analysis are treated as validation perspectives rather than separate methodological contributions.

During Check Point 3, the RDRA mechanism was formalised, its distinction from Equal Borda was made explicit, the five deployment dimensions were clearly scoped, VQ1 criterion-uncertainty results were subjected to paired statistical validation, reliability-weight sensitivity was examined, VQ2 responsiveness was evaluated under controlled AI4I sensor corruption, VQ3 examined downstream usefulness under changing stakeholder preferences, and complementary evidence was obtained from NASA C-MAPSS FD001 [6].

The current Main Objective:

To develop and empirically validate a reliability-aware rank-aggregation methodology for deployment-oriented predictive-maintenance model selection that accounts for differences in the stability of evaluation evidence and produces a defensible consensus ranking under evaluation uncertainty.

The current research is organised around the following Main Research Question:

How can reliability-aware rank aggregation produce a robust and defensible consensus ranking of predictive-maintenance model configurations when deployment criteria provide conflicting and differently stable empirical evidence?

Three complementary validation questions are used:

- **VQ1 - Criterion Uncertainty:** Does RDRA provide stronger consensus consistency when the available deployment criteria vary?
- **VQ2 - Data-Quality Uncertainty:** Does the ranking remain appropriately responsive when meaningful changes occur in the underlying deployment evidence?
- **VQ3 - Preference Uncertainty:** Does the resulting consensus support useful downstream deployment decisions when stakeholder priorities vary?

These validation questions aim to evaluate the ranking and decision methodology.

The current evidence supports a deliberately bounded conclusion. Primary RDRA shows statistically supported improvements over Equal Borda on several VQ1 ranking-consistency measures. Besides, a separately calibrated and locked exploratory refined-RDRA configuration shows statistically stronger VQ2 responsiveness under the tested AI4I data-quality perturbations. However, VQ3 does not currently show an additional Top-3 downstream decision-usefulness advantage over Equal Borda.

Hence, the project remains on track. The central methodology and major validation framework are substantially established. The remaining work concerns integrated results interpretation, literature and novelty positioning, final sensitivity checking, publication-oriented writing, additional supervisor feedback and final thesis/research communication.

2. Weekly Research Progress Diary

The following diary summarises the weekly research progress, key evidence, problems identified during supervision and the corrective actions taken.

Table 2.1 Weekly Research Progress Diary

Week	Meeting date	Main progress	Key evidence	Impediment / supervisor feedback	Action taken
Week 6	21/08/2026	Consolidated RDRA as the central methodological contribution; clarified Main RQ and VQ1-VQ3; formalised criterion-ranking reliability estimation and aggregation; clarified AI4I and C-MAPSS experimental roles.	Figures 1-5	Contribution needed to be distinguished clearly from established Borda/rank aggregation. "Reliability" could also be misunderstood as stakeholder preference.	Defined reliability as empirical criterion-ranking reproducibility; separated established aggregation components from the proposed reliability-aware deployment use.
Week 7	28/08/2026	Completed primary VQ1 evaluation and paired statistical validation; completed reliability-weight sensitivity checking; presented updated methodology and	Figures 6-8	Supervisor requested clearer explanation of why consensus ranking is needed, what specifically differs in RDRA, how the five dimensions are scoped, and why ranking	Added Equal Borda-RDRA comparison; clarified dimension scope; strengthened sensitivity interpretation; developed VQ2 responsiveness and VQ3 downstream decision analysis.

		validation logic to supervisor.		stability alone is insufficient.	
Week 8	04/09/2026	Completed locked refined-RDRA VQ2 comparison, VQ3 decision-usefulness analysis, complementary C-MAPSS evaluation and consolidated VQ1-VQ3 interpretation; progressed publication positioning.	Figures 9-15	Final claims must follow the evidence rather than imply universal superiority. Exploratory refinement must remain separate from primary RDRA. VQ3 does not provide an additional Top-3 advantage.	Preserved primary $\gamma = 1$ RDRA; labelled calibrated method as an exploratory refinement; reported neutral VQ3 evidence transparently; narrowed contribution claim.

This diary demonstrates the Check Point 3 requirements that research progress is being monitored continuously, evidence of milestone completion is retained, and research problems are identified early and brought to the supervisor's supervision for corrective action.

3. Week 6 - Methodological Consolidation and Formalisation

3.1 Research Problem and Decision Setting

Predictive-maintenance research has developed extensive methods for fault diagnosis, failure prediction and RUL estimation [1]-[4]. However, industrial deployment introduces additional considerations beyond predictive performance. Model-selection research similarly demonstrates that some well-performed algorithm can depend on task characteristics, evaluation objectives and selection strategy [5], [14]. Thus, the best predictive model is not necessarily the best deployment choice.

The present AI4I evaluation operationalises five deployment-oriented dimensions.

Table 3.1 Five deployment-oriented dimensions

Deployment dimension	Experimental evidence	Deployment purpose
Predictive Quality	F1-score and PR-AUC	Measures predictive discrimination and minority-class failure-prediction performance
Robustness	Performance under controlled sensor corruption	Evaluates resilience when input evidence is degraded
Probability Reliability	Brier score and Expected Calibration Error	Evaluates whether predicted probabilities are sufficiently trustworthy
Computational Efficiency	Inference latency and model size	Represents computational and deployment overhead
Operational Impact	False-negative and false-positive cost scenarios	Represents asymmetric consequences of prediction errors

These five dimensions are the dimensions operationalised in this research. Robustness is especially relevant in predictive-maintenance environments because real sensing systems may be affected by noise, missing measurements, outliers or sensor degradation [9]. Probability reliability is also crucial because discrimination performance alone does not establish whether predicted probabilities correspond appropriately to observed risk. Moreover, calibration is a different property of probabilistic predictive systems [10]. Operational decision-making can additionally involve non-equivalent fault costs and asymmetric consequences [11].

.

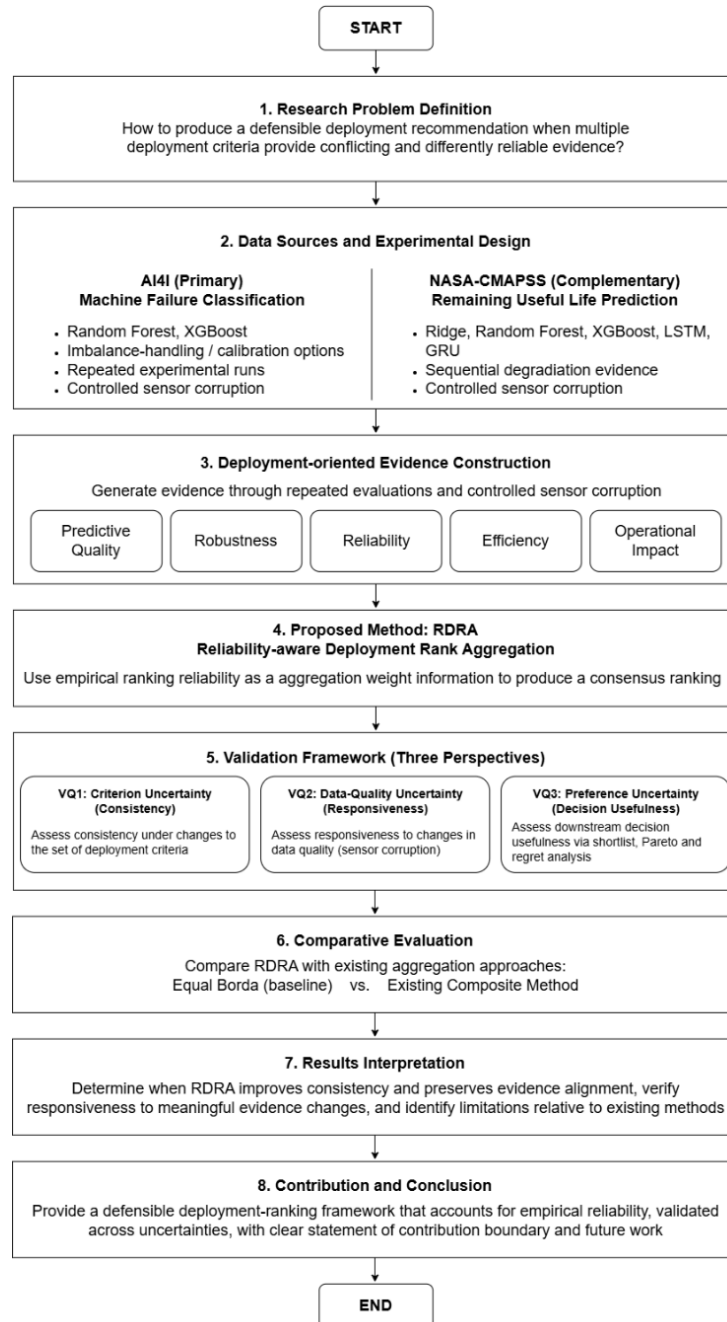


Figure 3.1 Overall research methodology for RDRA development and validation

Figure 3.1 summarises the overall research workflow developed during this checkpoint, linking the research problem, experimental evidence, RDRA aggregation, VQ1-VQ3 validation, baseline comparison and final interpretation. This figure provides evidence that the project has progressed from an initial research idea into a structured and testable methodology. It is particularly important because it shows how the three validation perspectives work together: VQ1 evaluates consistency, VQ2 evaluates responsiveness, and VQ3 evaluates downstream decision usefulness.

3.2 Experimental Datasets and Their Roles

The research uses two predictive-maintenance datasets with deliberately different methodological roles.

AI4I 2020

The AI4I 2020 Predictive Maintenance Dataset is a synthetic benchmark designed to reflect characteristics of industrial predictive-maintenance data [8]. It is used as the primary RDRA development and validation environment because the experimental pipeline offers the repeated evidence needed to estimate criterion-specific ranking reproducibility.

AI4I supports the full set of five deployment dimensions through predictive-performance, robustness, calibration, computational and operational-cost evidence.

NASA C-MAPSS FD001

NASA's Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) turbofan dataset contains simulated run-to-failure trajectories and multivariate sensor measurements generated for prognostics research [20], [21]. NASA states that four dataset groups were generated under different combinations of operational conditions and fault modes using C-MAPSS.

The current project uses FD001 as a complementary RUL-regression setting. Its experimental evidence supports predictive, robustness and computational-efficiency analysis.

C-MAPSS is not presented as a second complete RDRA validation, because the current C-MAPSS pipeline does not provide the same repeated criterion-level evidence required to estimate the complete set of RDRA reliability weights [7]. Instead, it offers complementary cross-task evidence concerning deployment-rank behaviour under a substantially different predictive-maintenance problem.

This distinction avoids the external-validation scope from being overstated.

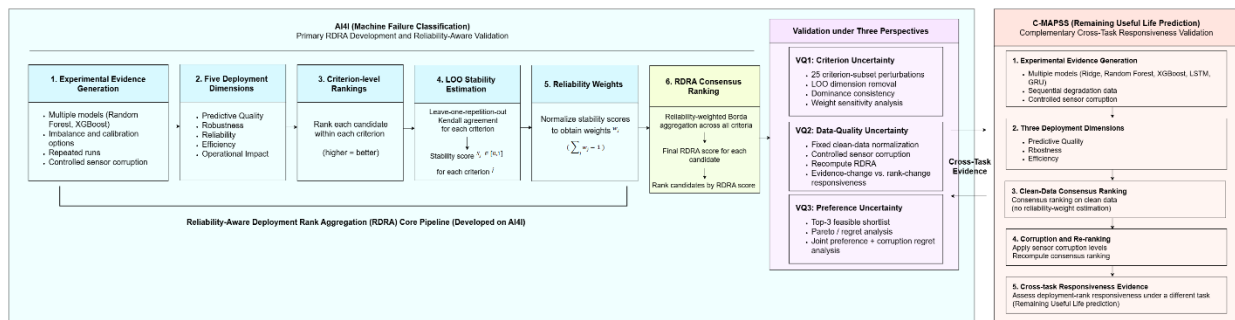


Figure 3.2 Detailed experimental framework showing AI4I primary evaluation and C-MAPSS complementary evaluation.

Figure 3.2 provides the detailed experimental framework implemented during this checkpoint. AI4I is used as the primary dataset for full RDRA development and validation because it provides repeated criterion-level evidence for estimating reliability weights. C-MAPSS FD001 is used as complementary cross-task evidence for deployment-rank responsiveness in an RUL prediction setting. This distinction is important because it demonstrates broader experimental coverage without overstating C-MAPSS as a second complete RDRA validation.

3.3 Defining a Defensible Deployment Ranking

Multi-criteria decision-making and aggregation research demonstrates that alternative rankings can arise when several criteria or methods are considered simultaneously [12], [13], [15]. Rank aggregation provides mechanisms for constructing consensus from such conflicting orderings [15], [18].

However, a central conceptual insight during Check Point 3 was that a defensible deployment ranking should not be judged only according to whether it remains unchanged. A ranking method that never changes could appear maximally stable even when its supporting evidence changes substantially.

Thus, the study defines four desirable properties:

1. **Evidence alignment:** the consensus should remain consistent with meaningful deployment evidence.
2. **Reliability awareness:** differences in empirical reproducibility between criterion rankings should be represented.
3. **Consistency:** moderate changes in criterion availability should not create disproportionate instability.
4. **Responsiveness:** meaningful changes in the underlying evidence should still be capable of changing the ranking.

This conceptual framework motivates VQ1 and VQ2 as complementary validation perspectives rather than redundant stability experiments.

3.4 Formalisation of Primary RDRA

3.4.1 Empirical Reliability

The meaning of reliability within RDRA was formally clarified during this checkpoint. It does not represent subjective stakeholder preference, business priority, importance assigned by an expert, universal intrinsic importance of a criterion.

Instead, RDRA operationalises reliability as the empirical reproducibility of a criterion-specific candidate ranking across repeated experimental evidence. This is crucial because weighted

MCDM frameworks can use weights to represent importance or preference [12], [13], whereas the weights in RDRA serve a different purpose. Therefore, an empirical reliability weight reflects evidence reproducibility, whereas a stakeholder-preference weight reflects decision priorities, they are not equivalent.

A criterion can be empirically reproducible but relatively unimportant to one stakeholder. Conversely, an important stakeholder criterion can be experimentally unstable. Hence, VQ3 treats stakeholder preference as a separate downstream decision layer.

3.4.2 Borda Transformation and Baseline

For candidate i and deployment dimension j , the criterion-level rank is represented by:

$$r_{ij}$$

The rank is converted to a normalised Borda score:

$$B_{ij} = \frac{n - r_{ij}}{n - 1}$$

where n is the number of candidate configurations.

The highest-ranked candidate receives:

$$B_{ij} = 1$$

while the lowest-ranked receives:

$$B_{ij} = 0$$

Equal Borda gives every criterion equal influence:

$$w_j^B = \frac{1}{m}$$

where m is the number of deployment dimensions.

The Equal Borda consensus score is:

$$S_i^B = \sum_{j=1}^m w_j^B B_{ij}$$

Rank aggregation and consensus construction are established research areas [15], [16].

3.4.3 Reliability Estimation

Purpose of the Comparison

RDRA adds an empirical reliability-estimation stage. For each deployment dimension, Leave-One-Repetition-Out (LOO) agreement is calculated between:

- the candidate ranking obtained from a held-out experimental repetition
- the reference ranking derived from the remaining repetitions

Kendall rank agreement is used to quantify ranking reproducibility [17].

For criterion j :

$$s_j = \frac{1}{R} \sum_{r=1}^R \max(0, \tau_{jr})$$

where:

- R is the number of repeated experiments
- τ_{jr} is the LOO Kendall agreement for repetition r

Negative agreement values are clipped to zero before averaging because they do not represent positive reproducibility evidence.

The resulting stability values are normalised:

$$w_j = \frac{s_j}{\sum_{k=1}^m s_k}$$

The primary RDRA consensus score is then:

$$S_i^{RDRA} = \sum_{j=1}^m w_j B_{ij}$$

Hence, the RDRA process begins by generating repeated experimental evidence, converting it into criterion-specific rankings, assessing the reproducibility of those rankings to obtain empirical reliability weights, transforming the ranks into normalised Borda scores, and finally aggregating them into a reliability-aware consensus ranking. The algorithm workflow of RDRA is shown in Figure 3.

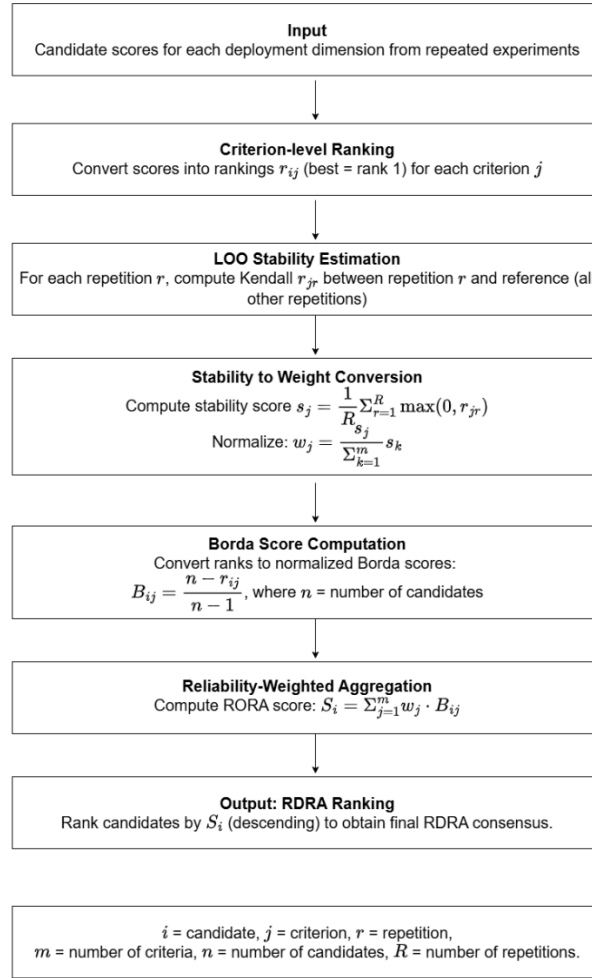


Figure 3.3 Algorithmic workflow of RDRA

Figure 3.3 illustrates the core RDRA workflow implemented in this study. Candidate scores from repeated experiments are first converted into criterion-specific rankings. Subsequently, Leave-One-Repetition-Out Kendall agreement is used to estimate the empirical reproducibility of each criterion ranking, which is converted into normalised reliability weights. These weights are combined with normalised Borda scores to calculate the final reliability-weighted consensus score and RDRA ranking.

This figure provides evidence that the proposed RDRA method has been formally specified during this checkpoint rather than remaining only a conceptual idea. It also makes the key distinction from Equal Borda explicit: RDRA estimates criterion reliability from repeated experimental evidence and uses this reliability information during aggregation, whereas Equal Borda gives every criterion equal influence.

	Variant	Existing_Composite_Score	Existing_Composite_Rank	Equal_Borda_Score	Equal_Borda_Rank	RDRA_Score	RDRA_Rank
14	XGBoost none platt	0.811233	1.0	0.788235	2.0	0.847071	1.0
13	XGBoost none none	0.808227	2.0	0.788235	1.0	0.842930	2.0
12	XGBoost none isotonic	0.791357	3.0	0.752941	3.0	0.806502	3.0
9	XGBoost class_weight isotonic	0.725130	5.0	0.705802	4.0	0.714351	4.0
11	XGBoost class_weight platt	0.744130	4.0	0.658024	5.0	0.648885	5.0
17	XGBoost smote platt	0.619150	7.0	0.494118	8.0	0.580567	6.0
10	XGBoost class_weight none	0.684466	6.0	0.600000	6.0	0.575520	7.0
15	XGBoost smote isotonic	0.588033	8.0	0.458824	9.5	0.541190	8.0
3	Random Forest none isotonic	0.556127	11.0	0.505882	7.0	0.493432	9.0
5	Random Forest none platt	0.564393	10.0	0.458824	9.5	0.465890	10.0
4	Random Forest none none	0.548208	12.0	0.411765	12.0	0.392798	11.0
16	XGBoost smote none	0.465335	15.0	0.364706	14.5	0.367593	12.0
0	Random Forest class_weight isotonic	0.539454	13.0	0.435294	11.0	0.384869	13.0
2	Random Forest class_weight platt	0.565439	9.0	0.388235	13.0	0.336787	14.0
1	Random Forest class_weight none	0.496431	14.0	0.364706	14.5	0.304788	15.0
8	Random Forest smote platt	0.405408	16.0	0.294118	17.0	0.265120	16.0
6	Random Forest smote isotonic	0.386370	17.0	0.294118	16.0	0.237438	17.0
7	Random Forest smote none	0.258919	18.0	0.235294	18.0	0.174269	18.0

Equal Borda weights: {'Predictive': np.float64(0.2), 'Robustness': np.float64(0.2), 'Reliability': np.float64(0.2), 'Efficiency': np.float64(0.2), 'Operational': np.float64(0.2)}

RDRA weights: {'Predictive': np.float64(0.2225604510066122), 'Robustness': np.float64(0.13016815637785187), 'Reliability': np.float64(0.2695485829128055), 'Efficiency': np.float64(0.27959794999654586), 'Operational': np.float64(0.0981249389698445)}

Figure 3.4 Main RDRA ranking and baselines

Figure 3.4 shows the final candidate rankings produced by primary RDRA, Equal Borda and the Existing Composite Method. RDRA uses empirically estimated reliability weights, whereas Equal Borda gives equal weight to all five deployment dimensions. The methods produce similar high-performing candidates but differ in some ranking positions, showing that reliability weighting can influence the final consensus.

This figure provides evidence that the RDRA method has been successfully implemented and produces a distinct deployment ranking for comparison with the baseline methods. The subsequent VQ1-VQ3 analyses are used to evaluate whether this ranking behaviour is more defensible under different uncertainty conditions.

3.5 Equal Borda versus RDRA

The distinction between Equal Borda and RDRA was made explicit following supervisor feedback.

Table 3.2 Comparison table of Equal Borda and RDRA

Stage	Equal Borda	RDRA
Criterion-specific ranking	r_{ij}	r_{ij}
Empirical reliability estimation	None	LOO Kendall ranking agreement
Criterion weight	$\frac{1}{m}$	$\frac{s_j}{\sum_k s_k}$
Borda transformation	Normalised Borda	Normalised Borda
Aggregation principle	Equal influence	Empirical reliability-weighted influence
Output	Consensus ranking	Reliability-aware consensus ranking

This comparison makes the contribution boundary clearer.

The project does not claim novelty in:

- rank aggregation
- Borda scoring
- Kendall agreement
- weighted multi-criteria decision-making
- existing robust rank-aggregation methodologies [12], [13], [15].

Instead, the proposed methodological adaptation is **the deployment-oriented use of repeated experimental ranking reproducibility as empirical reliability information when aggregating conflicting predictive-maintenance deployment rankings.**

The corresponding hypothesis is deliberately narrow, which is if deployment-criterion rankings differ in empirical reproducibility, incorporating that information may produce a more defensible consensus than imposing equal influence on all criterion rankings.

4. Week 7 - VQ1 Criterion-Uncertainty Validation

4.1 Purpose

VQ1 evaluates whether the consensus ranking remains consistent when the set of available deployment dimensions changes.

This represents criterion uncertainty: in practice, all evaluation evidence may not always be available, or different deployment analyses may emphasise different subsets of criteria.

With five dimensions, every subset containing two, three and four dimensions was evaluated:

$$\binom{5}{2} + \binom{5}{3} + \binom{5}{4} = 10 + 10 + 5 = 25$$

The complete five-dimension ranking was retained as the reference and was not counted as an additional perturbation.

Across the 25 matched criterion-subset perturbations, primary RDRA achieved:

- higher mean Kendall agreement
- higher mean Spearman agreement
- greater mean Top-3 overlap
- lower mean absolute rank displacement

than Equal Borda.

The VQ1 output also included an evidence-alignment/dominance check. Nine comparable dominance relationships were identified and no dominance violations occurred, providing an

additional sanity check that the resulting consensus was not contradicting the observed deployment evidence.

Evidence 1 - VQ1 comparison of Equal Borda, Existing Composite and RDRA across 25 criterion-subset perturbations

	Method	Mean_Kendall	Mean_Spearman	Mean_Top3_Overlap	Mean_Absolute_Rank_Displacement	Tie_Aware_Winner_Survival	Mean_Top1_Set_Overlap	N_Perturbations
0	Equal_Borda	0.677306	0.799260	0.720000	2.257778	0.32	0.20	25
1	Existing_Composite	0.624052	0.741754	0.746667	2.448889	0.36	0.36	25
2	RDRA	0.746405	0.862456	0.813333	1.773333	0.36	0.36	25

Figure 4.1 VQ1 summary statistics across 25 criterion-subset perturbations

The output summarises the consistency results for Equal Borda, Existing Composite and RDRA across 25 matched criterion-subset perturbations. RDRA achieved the highest mean Kendall agreement (0.746), mean Spearman agreement (0.862) and mean Top-3 overlap (0.813), while also producing the lowest mean absolute rank displacement (1.773).

This provides initial evidence that RDRA produced a more consistent consensus ranking than the baseline approaches when the available deployment criteria changed.

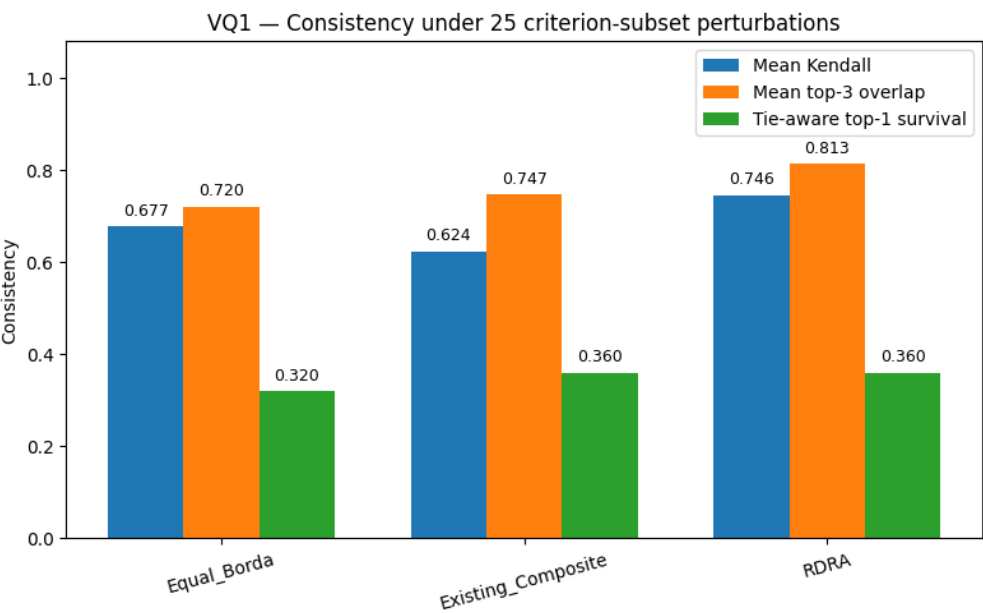


Figure 4.2 VQ1 consistency of Existing Composite, Equal Borda and primary RDRA

The figure visualises the main VQ1 consistency measures for the three aggregation methods. Based on the result, RDRA shows the strongest mean Kendall agreement and Top-3 preservation, while tie-aware Top-1 survival remains similar to the Existing Composite Method.

This visual comparison supports the descriptive VQ1 finding that RDRA is generally more stable under criterion uncertainty, while the following statistical-validation analysis is used to determine which improvements are statistically supported.

4.2 VQ1 Statistical Validation

Because RDRA and Equal Borda were evaluated on the same 25 criterion subsets, the observations form matched pairs.

The statistical analysis therefore included:

- paired t -tests
- 95% confidence intervals
- paired Cohen's d_z
- Shapiro-Wilk diagnostics
- Wilcoxon signed-rank robustness checks
- Holm correction across the four principal VQ1 comparisons

Holm's sequentially rejective procedure was applied to control the family-wise error rate associated with multiple comparisons [19].

The principal VQ1 results were:

Table 4.1 VQ1 paired statistical-validation results

Metric	Mean RDRA improvement	Interpretation
Kendall agreement	+0.069	Statistically supported after Holm correction
Spearman agreement	+0.063	Positive descriptive improvement but not significant after correction
Top-3 overlap	+0.093	Statistically supported after Holm correction
Mean absolute rank displacement	approximately 0.484 ranks better	Statistically supported after Holm correction

Therefore, the conclusion we get from the VQ1 statistical validation is primary RDRA demonstrates statistically supported improvements over Equal Borda on several measures of ranking and shortlist consistency under criterion uncertainty.

4.2.1 Evidence of Achievement

Evidence 2 - VQ1 paired statistical-validation results

	Metric	Mean improvement	95% CI	Holm-adjusted paired t-test	Effect
0	Kendall	+0.069	[0.012, 0.126]	0.039	Moderate
1	Spearman	+0.063	[-0.007, 0.134]	0.077	Small
2	Top-3 overlap	+0.093	[0.030, 0.156]	0.022	Moderate
3	Rank displacement	+0.484 better	[0.146, 0.822]	0.022	Moderate

Figure 4.3 VQ1 paired statistical-validation summary

The output summarises the paired statistical comparison between RDRA and Equal Borda across the 25 matched criterion-subset perturbations. RDRA achieved statistically supported improvements in Kendall agreement (+0.069), Top-3 overlap (+0.093) and rank displacement (+0.484 ranks better) after Holm correction. The Spearman improvement (+0.063) was positive but not statistically significant after correction.

This provides stronger evidence that the VQ1 consistency improvements are not only descriptive but are statistically supported on several important measures.

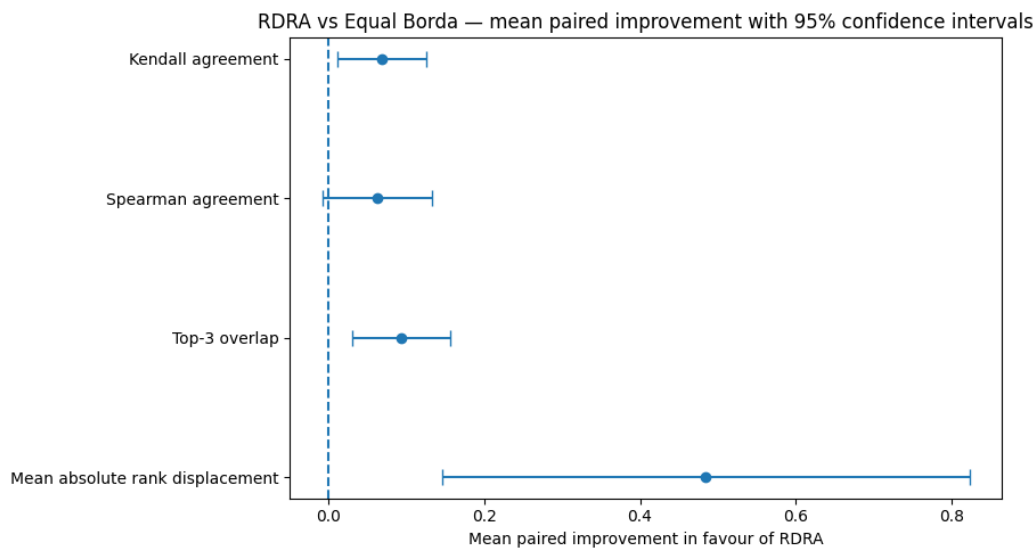


Figure 4.4 Mean paired RDRA improvement over Equal Borda with 95% confidence interval

The output visualises the mean paired improvements and their 95% confidence intervals. The intervals for Kendall agreement, Top-3 overlap and rank displacement remain above zero, while the Spearman interval crosses zero. This supports the bounded VQ1 result that, on some but not all evaluated measures, RDRA exhibits statistically stronger consistency than Equal Borda.

4.3 Reliability-Weight Sensitivity Analysis

A further methodological concern was whether the RDRA consensus depended excessively on one exact estimate of the empirical reliability weights.

To investigate this issue, 1,000 controlled reliability-weight perturbations were performed.

The resulting statistics were:

Table 4.2 RDRA reliability-weight sensitivity results

Measure	Result	Interpretation
Mean Kendall agreement	0.964	Strong preservation of overall ranking structure
Median Kendall agreement	0.974	High typical agreement with the original RDRA ranking
Mean Top-3 overlap	1.000	Complete preservation of the Top-3 candidate set
Mean absolute rank displacement	0.296	Only minor candidate movement
Tie-aware Top-1 survival	0.699	Exact winner is more sensitive than the broader ranking

This experiment evaluates the internal sensitivity of RDRA to uncertainty in its estimated reliability weights. The results indicate that the wider ranking structure and Top-3 shortlist are highly stable under the tested perturbations, while the exact identity of the first-ranked candidate retains greater sensitivity.

4.3.1 Evidence of Achievement

Evidence 3 - RDRA reliability-weight sensitivity results across 1,000 controlled perturbations

RDRA weight-estimation sensitivity								
N_Runs	Perturbation_SD	Mean_Kendall	Median_Kendall	Mean_Top3_Overlap	Mean_Rank_Displacement	Tie_Aware_Top1_Survival	Mean_Top1_Set_Overlap	
0	1000	0.05	0.964261	0.973856	1.0	0.296444	0.699	0.699

Figure 4.5 Summary of RDRA reliability-weight sensitivity across 1,000 perturbations

The output summarises the sensitivity analysis performed by perturbing the estimated RDRA reliability weights across 1,000 runs. The results show high mean and median Kendall agreement (0.964 and 0.974), complete mean Top-3 preservation (1.000), low mean rank displacement (0.296) and a tie-aware Top-1 survival rate of 0.699.

This offers evidence that the broader RDRA ranking and Top-3 shortlist remain highly stable under small changes to the estimated reliability weights, although the exact first-ranked candidate is more sensitive.

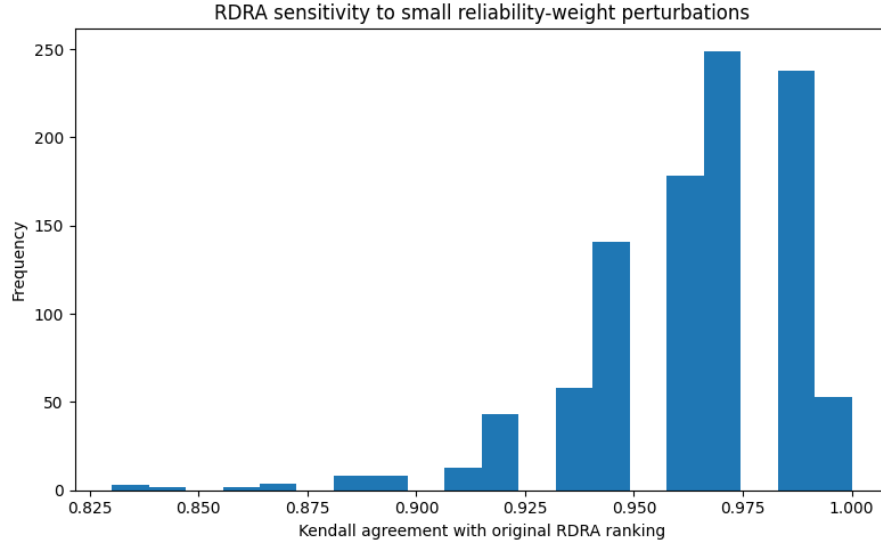


Figure 4.6 Distribution of Kendall agreement under reliability-weight perturbations

The output shows the distribution of Kendall agreement between the perturbed rankings and the original RDRA ranking. Therefore, most perturbations produce very high agreement, with the distribution concentrated close to 1.0.

This visually supports the sensitivity result that the overall RDRA ranking is robust to small uncertainty in the estimated reliability weights, while avoiding a claim that every individual ranking position is unchanged.

4.4 Exploratory Reliability-Influence Refinement

During Check Point 3, an additional controlled refinement was investigated without changing the supervisor-approved primary RDRA methodology.

Primary RDRA remains:

$$w_j = \frac{s_j}{\sum_k s_k}$$

The exploratory extension introduces a reliability-influence parameter:

$$w_j(\gamma) = \frac{s_j^\gamma}{\sum_k s_k^\gamma}$$

This gives the following interpretation:

$$\begin{aligned} \gamma = 0 &\Rightarrow \text{Equal Borda,} \\ \gamma = 1 &\Rightarrow \text{Primary RDRA,} \end{aligned}$$

While $\gamma = 3.0$ strengthens the influence of differences in empirical reliability.

To reduce the risk of validation leakage, the γ search was conducted using a development-only subset of VQ1 criterion perturbations. The selected value was then locked before its subsequent held-out and VQ2 use.

Within the tested grid, the selected value was $\gamma = 3.0$. However, this value occurred at the upper boundary of the tested grid. Therefore, $\gamma = 3.0$ not be described as a globally optimal value.

The correct terminology is locked exploratory refined RDRA. The primary methodology remains the supervisor-approved $\gamma = 1$.

5. Week 8 - VQ2 Data-Quality Responsiveness

5.1 Motivation

One of the most important conceptual issues identified through supervision was that stability alone cannot define ranking quality.

A ranking algorithm that never changes would appear perfectly stable even if the evidence supporting the candidate models changed substantially. Therefore, a defensible consensus must balance two behaviours: resisting unnecessary change while remaining responsive to meaningful evidence change. Hence, VQ1 evaluates the first property. VQ2 evaluates the second.

5.2 Controlled Data-Quality Perturbations

The primary VQ2 AI4I analysis evaluates 16 controlled corruption scenarios from four corruption families:

1. Gaussian noise
2. Missing Completely at Random (MCAR) missingness
3. Outliers
4. static sensor offset

Multiple severity levels are applied within these families.

The evaluation uses fixed clean-data normalisation so that the reference evaluation scale remains fixed while the sensor evidence is deliberately degraded. This is crucial because changing the normalisation reference simultaneously with the corrupted evidence could obscure the magnitude of the underlying deployment-evidence change. Robustness to sensor degradation is especially relevant to predictive-maintenance systems because real-world sensing environments can contain noise and degraded measurements [9].

5.2.1 Evidence of Achievement

Evidence 4 - VQ2 locked refined-RDRA responsiveness under controlled AI4I sensor corruption

	Condition	Level	Evidence_Change_Magnitude	Refined_RDRA_Rank_Displacement	Equal_Borda_Rank_Displacement
0	Gaussian Noise	0.05	0.227783	0.333333	0.888889
1	Gaussian Noise	0.10	0.301766	0.555556	1.611111
2	Gaussian Noise	0.20	0.321858	0.888889	2.333333
3	Gaussian Noise	0.30	0.305759	0.777778	2.333333
4	Missing Values MCAR	0.05	0.269785	0.333333	1.722222
5	Missing Values MCAR	0.10	0.307266	0.555556	2.055556
6	Missing Values MCAR	0.20	0.327334	1.222222	3.000000
7	Missing Values MCAR	0.30	0.322597	0.888889	3.000000
8	Outliers	0.01	0.300420	0.555556	2.388889
9	Outliers	0.03	0.313063	0.888889	3.166667
10	Outliers	0.05	0.298197	0.888889	3.277778
11	Outliers	0.10	0.286202	0.777778	3.222222
12	Static Sensor Offset	0.05	0.267268	0.333333	1.111111
13	Static Sensor Offset	0.10	0.311245	0.333333	0.777778
14	Static Sensor Offset	0.20	0.307642	0.888889	2.500000
15	Static Sensor Offset	0.30	0.296094	0.777778	2.111111

Figure 5.1 VQ2 evidence-change and rank-displacement results across 16 AI4I corruption scenarios

The output summarises the 16 controlled AI4I corruption scenarios and records the magnitude of deployment-evidence change together with the corresponding rank displacement for locked refined RDRA and Equal Borda.

This provides direct evidence that the VQ2 responsiveness analysis was completed across Gaussian noise, MCAR missingness, outliers and static sensor offset at multiple severity levels.

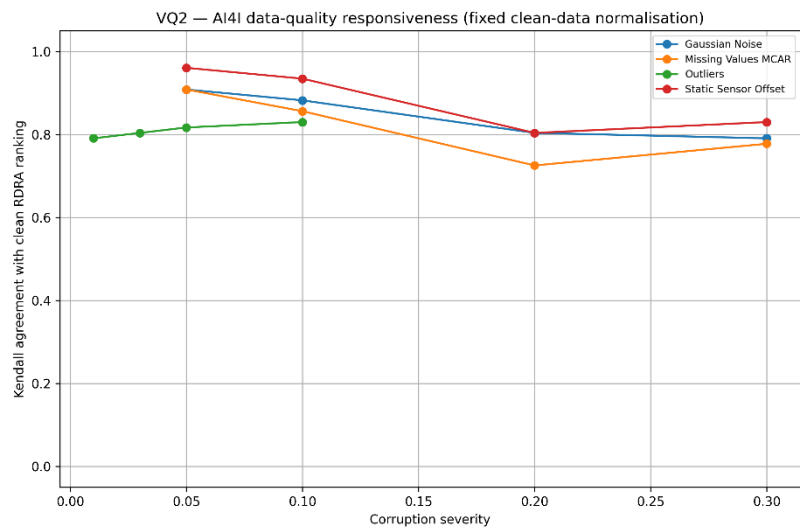


Figure 5.2 VQ2 ranking consistency under increasing AI4I corruption severity

The figure shows how agreement with the clean-data RDRA ranking changes across increasing corruption severity for the four corruption families. Ranking agreement generally remains relatively high, although the amount of change differs by corruption type and severity.

This figure demonstrates that the method is not simply unstable under corrupted data, while also showing that meaningful sensor degradation can alter the resulting ranking.

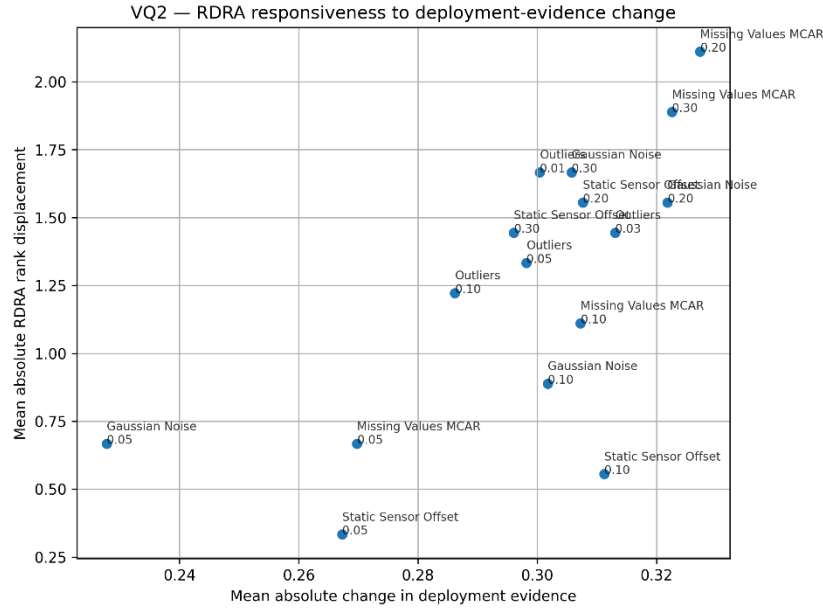


Figure 5.3 VQ2 refined-RDRA responsiveness to deployment-evidence change

The figure plots the magnitude of deployment-evidence change against the corresponding refined-RDRA rank displacement across the 16 corruption scenarios. The overall positive relationship indicates that larger changes in deployment evidence tend to be associated with larger changes in the consensus ranking.

This offers the main descriptive evidence for responsiveness, showing that the ranking can react when its supporting deployment evidence changes. The following VQ2 statistical comparison is then used to determine whether this responsiveness is stronger than Equal Borda.

5.3 VQ2 Responsiveness Measure

For each corruption scenario, define:

$$\Delta E = \text{deployment evidence-change magnitude}$$

and:

$$\Delta R = \text{ranking-change magnitude}$$

Responsiveness is evaluated through Spearman association:

$$\rho = Corr_s(\Delta, E\Delta R)$$

A stronger positive association means that larger changes in deployment evidence tend to correspond to larger changes in the resulting consensus ranking.

For the locked exploratory refined RDRA: $\rho_{\text{Refined RDRA}} = 0.678$ with $p = 0.0039$.

For Equal Borda: $\rho_{\text{Equal Borda}} = 0.348$ with $p = 0.187$.

However, it would be statistically incorrect to infer superiority merely because one individual correlation has $p < 0.05$ and the other does not.

5.3.1 Evidence of Achievement

Evidence 5 - Locked exploratory refined RDRA versus Equal Borda responsiveness comparison

	Method	Spearman_Rho	P_Value_Two_Sided	Bootstrap_CI_Lower	Bootstrap_CI_Upper	N_Scenarios	N_Bootstrap_Valid
0	Refined RDRA	0.678041	0.003892	0.180088	0.937348	16	10000
1	Equal Borda	0.347570	0.187141	-0.235391	0.825839	16	10000
	RDRA_Spearman	Equal_Borda_Spearman	Difference_RDRA_minus_Borda	Bootstrap_CI95_Lower	Bootstrap_CI95_Upper	N_Scenarios	
0	0.678041	0.34757	0.330471	0.046026	0.702235	16	

Figure 5.4 VQ2 responsiveness statistics for locked refined RDRA and Equal Borda

The output reports the responsiveness correlations for the 16 AI4I corruption scenarios. Locked refined RDRA achieved a Spearman responsiveness of 0.678, compared with 0.348 for Equal Borda. The direct difference was $\Delta\rho = +0.330$, with a matched-bootstrap 95% confidence interval of [0.046, 0.702].

This provides detailed statistical evidence used to compare the responsiveness of the two aggregation approaches.

	Metric	Refined RDRA	Equal Borda	Difference ($\Delta\rho$)	95% CI	RDRA p-value	Equal Borda p-value
0	Spearman responsiveness (ρ)	0.678041	0.34757	0.330471	[0.046, 0.702]	0.003892	0.187141

Figure 5.5 Direct statistical comparison of VQ2 responsiveness

The output summarises the direct comparison between locked refined RDRA and Equal Borda. The positive difference of +0.330 and the 95% confidence interval CI [0.046, 0.702], which remains above zero, provides statistical support for stronger refined-RDRA responsiveness under the tested corruption scenarios. This is the key inferential evidence for the VQ2 comparison.

	VQ2 statistical result	Refined RDRA	Equal Borda
0	Spearman responsiveness ρ	0.678	0.348
1	Individual two-sided p	0.0039	0.187
2	Direct difference $\Delta\rho$	+0.330	—
3	Bootstrap 95% CI for $\Delta\rho$	[0.046, 0.702]	—

Figure 5.6 Consolidated VQ2 statistical-validation result

The output presents the main VQ2 findings in a concise form. Locked refined RDRA achieved higher responsiveness than Equal Borda, while the matched-bootstrap confidence interval for the direct difference excludes zero.

This offers a clear checkpoint summary that the locked exploratory refined RDRA demonstrated statistically stronger responsiveness than Equal Borda under the tested AI4I data-quality perturbations.

5.4 VQ2 Direct Statistical Validation

The direct responsiveness difference is:

$$\begin{aligned}\Delta\rho &= \rho_{\text{Refined RDRA}} - \rho_{\text{Equal Borda}} \\ &= 0.678 - 0.348 = +0.330.\end{aligned}$$

A matched bootstrap analysis using 10,000 resamples was applied to estimate uncertainty in this direct method difference. Bootstrap resampling provides a well-established approach for estimating uncertainty without requiring a closed-form sampling distribution for every statistic [18].

The resulting interval was: 95% $CI = [0.046, 0.702]$.

Because the entire interval lies above zero, the result supports the conclusion that the locked exploratory refined RDRA exhibits a stronger evidence-change/ranking-change association than Equal Borda under the tested AI4I data-quality perturbations.

Therefore, the conclusion is Under the 16 tested AI4I corruption scenarios, the locked exploratory refined RDRA demonstrates statistically stronger responsiveness to changes in deployment evidence than Equal Borda, based on the direct matched-bootstrap comparison.

This complements VQ1:

VQ1: resist unnecessary instability

Whereas VQ2:

VQ2: respond to meaningful evidence change

Therefore, VQ1 and VQ2 provide a more defensible validation framework than ranking stability alone.

5.4.1 Evidence of Achievement

Evidence 6 - Bootstrap 95% confidence interval for the direct responsiveness difference between locked exploratory refined RDRA and Equal Borda, $\Delta\rho$.

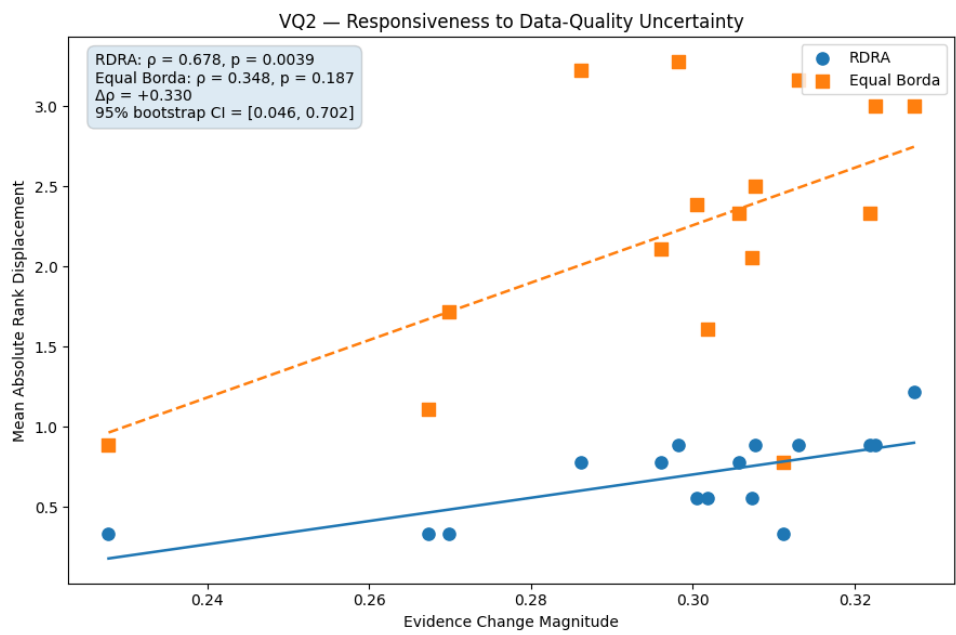


Figure 5.7 VQ2 responsiveness comparison between locked refined RDRA and Equal Bord

The figure compares the relationship between deployment-evidence change and rank displacement for locked refined RDRA and Equal Borda across the 16 corruption scenarios. Refined RDRA shows the stronger responsiveness correlation ($\rho = 0.678$) compared with Equal Borda ($\rho = 0.348$).

This provides visual evidence that refined-RDRA ranking changes are more closely aligned with changes in the underlying deployment evidence.

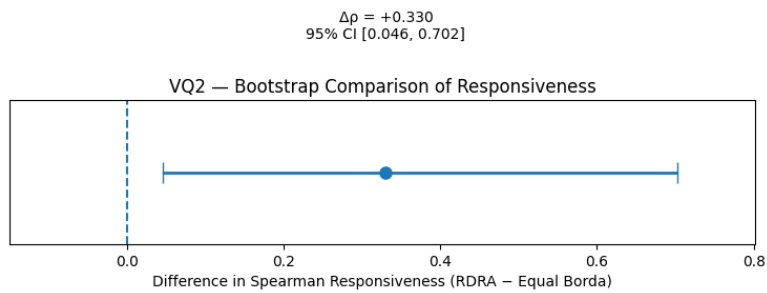


Figure 5.8 Bootstrap 95% confidence interval for the direct VQ2 responsiveness difference

The figure shows the matched-bootstrap estimate of the direct responsiveness difference between the two methods. The observed difference is $\Delta p = +0.330$, with a 95% confidence interval of [0.046, 0.702].

5.5 VQ3 - Preference Uncertainty and Decision Usefulness

VQ3 evaluates a different research property:

Does the consensus ranking remain useful for downstream deployment decisions when stakeholder preferences vary?

This distinction is necessary because objective empirical reliability weights should not be interpreted as replacing stakeholder priorities. Weighted multi-criteria research demonstrates that weighting assumptions can substantially affect decision aggregation [12], [13].

The VQ3 analysis includes:

- feasible Top-3 candidate shortlists
- 5,000 matched deployment-preference scenarios
- Pareto analysis
- scenario-level regret
- comparison with Equal Borda and Existing Composite
- joint preference-plus-corruption regret

The current results do not demonstrate an additional Top-3 downstream decision-usefulness advantage for RDRA over Equal Borda. Under the tested preference scenarios, their Top-3 shortlist regret outcomes are broadly comparable.

Therefore, no additional Top-3 downstream decision-usefulness advantage for RDRA has currently been demonstrated under the tested preference scenarios.

5.5.1 Evidence of Achievement

Evidence 7 - VQ3 feasible Top-3 shortlist and regret analysis across 5,000 matched deployment-preference scenarios

	Method	Top3_Feasible_Shortlist	Mean_Shortlist_Regret	P95_Shortlist_Regret	CVaR95_Shortlist_Regret	Worst_Case_Shortlist_Regret	Zero_Regret_Frequency
0	Equal_Borda	XGBoost class_weight isotonic XGBoost ...	0.002069	0.016802	0.038217	0.079113	0.9316
1	RDRA	XGBoost class_weight isotonic XGBoost ...	0.002069	0.016802	0.038217	0.079113	0.9316
2	Existing_Composite	XGBoost class_weight platt XGBoost cla...	0.003082	0.019930	0.057028	0.139577	0.9260

Figure 5.9 VQ3 feasible Top-3 shortlist and regret analysis across 5,000 matched deployment-preference scenarios

The output summarises the Top-3 shortlist and regret results for RDRA, Equal Borda and the Existing Composite Method across 5,000 matched preference scenarios. RDRA and Equal Borda

produce very similar shortlist-regret results, while the Existing Composite Method shows slightly higher regret on the reported measures.

This provides evidence that the VQ3 downstream decision-usefulness analysis was completed and shows that RDRA does not currently demonstrate an additional Top-3 decision-usefulness advantage over Equal Borda. Therefore, the result is used to define the current contribution boundary rather than to claim superiority.

Evidence 8 - VQ3 joint preference-plus-corruption regret analysis

	Variant	Mean_Joint_Regret	P95_Joint_Regret	Worst_Case_Joint_Regret	Zero_Regret_Frequency	Joint_Robust_Regret_Rank
3	XGBoost class_weight platt	0.123966	0.560538	0.969285	0.089765	1
2	XGBoost class_weight isotonic	0.084179	0.534891	0.971801	0.686618	2
0	Random Forest class_weight isotonic	0.533107	0.981915	0.998236	0.043500	3
1	Random Forest class_weight platt	0.530673	1.000000	1.000000	0.180118	4

Figure 5.10 VQ3 joint preference-plus-corruption regret analysis

The output summarises candidate-level regret when stakeholder-preference uncertainty and sensor corruption are considered jointly. The XGBoost configurations receive the two highest joint-robust ranks, while the Random Forest configurations show substantially higher regret under the combined uncertainty conditions.

This offers evidence that the VQ3 analysis was extended beyond preference uncertainty alone to examine candidate robustness under combined preference and data-quality uncertainty. The figure supports interpretation of which candidate configurations remain more defensible under joint uncertainty, rather than serving as a direct RDRA-versus-Equal-Borda comparison.

That final distinction is important for accuracy: Figure 20 evaluates candidate-level joint robustness, while Figure 19 is the stronger evidence for comparing the downstream Top-3 usefulness of RDRA and Equal Borda.

6. Complementary NASA C-MAPSS Cross-Task Evidence

The AI4I results constitute the primary RDRA development and validation evidence. To investigate whether the broader deployment-ranking problem remains relevant in a different predictive-maintenance task, complementary experiments were performed using NASA C-MAPSS FD001.

C-MAPSS was produced using the Commercial Modular Aero-Propulsion System Simulation to generate run-to-failure turbofan-engine sensor trajectories for prognostics research [20], [21]. The dataset has subsequently become a widely used benchmark for RUL prediction and machinery prognostics [6].

The present C-MAPSS experiment uses candidate RUL-prediction configurations and evaluates predictive performance, robustness under controlled corruption and computational efficiency.

Because the current C-MAPSS pipeline does not provide the same repeated criterion-level experimental structure required by the complete AI4I reliability-estimation procedure.

Therefore, is it claim that C-MAPSS provides complementary cross-task evidence concerning deployment-rank responsiveness in a different predictive-maintenance setting involving sequential degradation and RUL regression.

6.1 Evidence of Achievement

Evidence 9 - C-MAPSS complementary cross-task responsiveness

	Condition	Level	Kendall_DRS	Spearman_DRS	Tie_Aware_Top1_Survival	Scenario_Winner_Set	Clean_Winner_Set
0	Gaussian_Noise	0.05	0.200000	0.000000	False	Ridge	XGBoost
1	Gaussian_Noise	0.10	0.000000	-0.100000	False	Ridge	XGBoost
2	Gaussian_Noise	0.20	0.316228	0.205196	True	Ridge XGBoost	XGBoost
3	Gaussian_Noise	0.30	0.800000	0.900000	True	XGBoost	XGBoost
4	Gradual_Offset_Drift	0.05	0.000000	0.100000	True	XGBoost	XGBoost

Figure 6.1 C-MAPSS deployment-rank agreement under selected corruption scenarios

The output summarises how the C-MAPSS deployment ranking changes relative to the clean-data ranking under selected corruption conditions. It reports Kendall and Spearman agreement, Top-1 survival and changes in the scenario winner set.

This provides evidence that the complementary C-MAPSS analysis was implemented and that different corruption conditions can change the deployment ranking and winner stability.

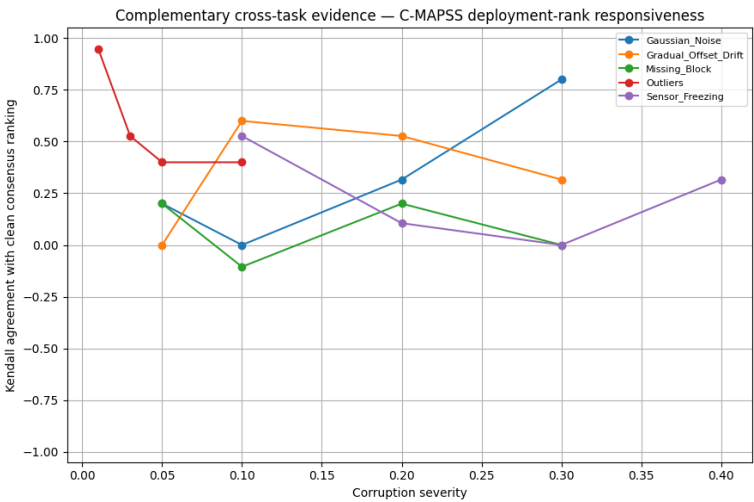


Figure 6.2 C-MAPSS complementary cross-task deployment-rank responsiveness

The figure shows Kendall agreement with the clean-data consensus ranking across different corruption types and severity levels. The ranking response varies across Gaussian noise, gradual offset drift, missing blocks, outliers and sensor freezing rather than following one identical pattern.

This provides complementary evidence that deployment rankings are sensitive to data-quality changes in a different predictive-maintenance setting involving RUL prediction, while not claiming that C-MAPSS constitutes a second full RDRA validation.

7. Consolidated VQ1-VQ3 Validation

The three validation perspectives answer related but distinct questions.

Table 7.1 VQ1-VQ3 Validation Summary

Validation perspective	Focus	Key evidence	Justification / interpretation	Supported conclusion
VQ1 - Criterion Uncertainty	Consistency	25 matched criterion-subset perturbations. Primary RDRA improved Kendall agreement by +0.069 , Top-3 overlap by +0.093 , and mean absolute rank displacement by approximately 0.484 ranks . These three measures remained statistically supported after Holm correction. Spearman improved by +0.063 , but was not Holm-significant.	Significant improvement in Kendall, Top-3 preservation and displacement indicates reduced sensitivity to changes in available criteria. The non-significant Spearman result prevents an overgeneralised claim that every consistency measure improves significantly.	Primary RDRA demonstrates stronger ranking and shortlist consistency than Equal Borda on several statistically supported measures under criterion uncertainty.
VQ2 - Data-Quality Uncertainty	Responsiveness	16 controlled AI4I corruption scenarios. Locked refined RDRA: $\rho = 0.678$; Equal	The confidence interval for the direct method difference lies entirely above zero. This	The locked exploratory refined RDRA demonstrates statistically stronger

		Borda: $\rho = 0.348$; $\Delta\rho = +0.330$; matched-bootstrap 95% CI [0.046, 0.702].	provides direct evidence that ranking change is more strongly aligned with deployment-evidence change for the locked exploratory refined method.	responsiveness than Equal Borda under the tested AI4I data-quality perturbations.
VQ3 - Preference Uncertainty	Decision usefulness	5,000 matched preference scenarios; feasible Top-3, Pareto, regret and joint preference-plus-corruption analyses. RDRA and Equal Borda produce broadly comparable current Top-3 regret outcomes.	Improved consistency or responsiveness does not necessarily produce better performance on every downstream decision metric. VQ3 therefore establishes a contribution boundary.	No additional Top-3 downstream decision-usefulness advantage for RDRA over Equal Borda has currently been demonstrated under the tested scenarios.

8. Corrective Action Following Supervisor Feedback

A major strength of this checkpoint is that supervisor feedback produced observable methodological changes.

Table 8.1 Corrective Action Following Supervisor Feedback

Supervisor feedback / identified problem	Corrective action	Result
Difference between Equal Borda and RDRA was insufficiently explicit.	Added stage-by-stage comparison.	Reliability-estimation and reliability-weighting stages are now visible.
Contribution/novelty required clearer definition.	Distinguished established Borda, rank aggregation and statistical components from the proposed deployment-oriented reliability mechanism.	Novelty boundary is narrower and more defensible.
Five deployment dimensions could appear exhaustive.	Defined them as the five dimensions operationalised within this study.	Avoids unsupported generalisation.
“Reliability” could be confused with stakeholder preference.	Defined reliability as empirical criterion-ranking reproducibility.	Reliability weighting and preference weighting are separated.
VQ1-VQ3 required clearer research roles.	Mapped VQ1 to consistency, VQ2 to responsiveness and VQ3 to downstream decision usefulness.	Validation framework now corresponds directly to desired ranking behaviour.
Stability alone does not establish ranking quality.	Added VQ2 evidence-change/ranking-change responsiveness.	Method is evaluated for both stability and meaningful responsiveness.
VQ1 descriptive differences required inferential evidence.	Added paired statistical tests, CIs, effect sizes, Holm correction and non-parametric checking.	Several VQ1 advantages now have statistical support.
Weight-sensitivity results required interpretation.	Added direction-specific interpretation for Kendall, Top-3 overlap, rank displacement and Top-1 survival.	Sensitivity evidence now explains what is stable and what remains sensitive.
Refinement could introduce validation leakage.	Used development-only VQ1 evidence for γ calibration and locked the result before VQ2/VQ3.	Primary RDRA remains unchanged; refinement is explicitly exploratory.

Method should not be described as universally superior.	Retained VQ3 neutral evidence and bounded the contribution.	
---	---	--

This progression directly demonstrates the Check Point 3 requirement that research impediments are identified and brought to the supervision sufficiently early for corrective action.

9. Milestones and Current Project Status

The methodology has now progressed from exploratory implementation into a form suitable for thesis and publication communication.

Table 9.1 Milestones and Current Project Status

Research milestone	Status
Initial literature review and research-gap identification	Completed
AI4I primary experimental framework	Completed
C-MAPSS complementary experimental framework	Completed
Deployment-oriented evidence construction	Completed
Primary RDRA algorithm development	Completed
Equal Borda baseline	Completed
Criterion-ranking empirical reliability estimation	Completed
Main RDRA ranking and baseline analysis	Completed
VQ1 criterion-subset validation	Completed
VQ1 paired statistical validation	Completed
Evidence-alignment/dominance sanity checking	Completed
Reliability-weight sensitivity analysis	Completed
Exploratory reliability-influence refinement	Completed
VQ2 AI4I data-quality responsiveness	Completed
VQ2 direct matched-bootstrap comparison	Completed
VQ3 preference uncertainty and decision usefulness	Completed
C-MAPSS complementary responsiveness analysis	Completed
Complete research methodology	Completed
Publication-venue evaluation	Completed
Similar research-paper identification	In progress
Integrated Results and Discussion	In progress
Publication-oriented paper preparation	Planned
Final thesis integration	Planned
Defence preparation	Planned

10. Evidence Achievement Summary

Table 10.1 Evidence Achievement Summary

Figure	Research achievement demonstrated
Figure 1	Overall RDRA development and validation methodology
Figure 2	AI4I primary evaluation and C-MAPSS complementary evaluation framework
Figure 3	Formal RDRA algorithmic workflow
Figure 4	Criterion-ranking stability estimates and empirical reliability weights
Figure 5	Main primary-RDRA ranking and aggregation baselines
Figure 6	VQ1 comparison across 25 criterion-subset perturbations
Figure 7	VQ1 paired statistical-validation results
Figure 8	Reliability-weight sensitivity across 1,000 controlled perturbations
Figure 9	Locked exploratory refined-RDRA VQ2 responsiveness
Figure 10	Locked refined RDRA versus Equal Borda VQ2 comparison
Figure 11	Matched-bootstrap CI for the direct VQ2 responsiveness difference
Figure 12	VQ3 feasible Top-3 shortlist and regret analysis
Figure 13	VQ3 joint preference-plus-corruption regret analysis
Figure 14	C-MAPSS complementary cross-task responsiveness
Figure 15	Consolidated VQ1-VQ3 validation summary

11. Current Research Contribution

The project proposes a **reliability-aware deployment rank-aggregation methodology for predictive-maintenance model selection**.

The methodological adaptation is:

to estimate the empirical reproducibility of criterion-specific candidate rankings from repeated experimental evidence and use this information as reliability weights when aggregating conflicting deployment-oriented rankings.

The current evidence supports three differentiated conclusions.

VQ1 - Consistency

Primary RDRA demonstrates statistically supported improvements over Equal Borda on several measures of ranking and shortlist consistency when the available deployment criteria vary.

VQ2 - Responsiveness

A leakage-controlled, locked exploratory refined-RDRA configuration demonstrates statistically stronger responsiveness than Equal Borda under the tested AI4I sensor-corruption scenarios.

VQ3 - Downstream Decision Usefulness

Current Top-3 decision usefulness remains broadly comparable to Equal Borda under the tested preference scenarios, and an additional RDRA Top-3 advantage has not been demonstrated.

Therefore, the current defensible contribution is:

RDRA incorporates empirically estimated criterion-ranking reproducibility as reliability information within deployment-oriented predictive-maintenance rank aggregation. Under the tested criterion-uncertainty conditions, primary RDRA demonstrates statistically supported improvements on several measures of ranking and shortlist consistency relative to Equal Borda. A separately calibrated and locked exploratory refinement demonstrates statistically stronger responsiveness to meaningful changes in deployment evidence under the tested AI4I data-quality perturbations. These ranking-level improvements do not currently establish a universal downstream decision-usefulness advantage.

12. Check Point 3 Reflection

Research Progress Check Point 3 represents an important transition from exploratory methodological development towards statistically supported and publication-oriented research.

Check Point 2 established RDRA as the central methodological direction and identified the need for clearer justification. During Check Point 3, the project addressed these issues by formally defining the RDRA mechanism, explicitly distinguishing it from Equal Borda, statistically validating VQ1 criterion-uncertainty performance, evaluating reliability-weight sensitivity, testing responsiveness under controlled sensor corruption, investigating downstream preference uncertainty and introducing a carefully separated exploratory reliability-influence refinement.

A major lesson from this period is that a proposed method should not be evaluated simply according to whether it produces numerically favourable outputs. The research must first define what constitutes desirable behaviour and then design experiments that directly test those properties.

This led to the complementary validation logic:

- VQ1 = consistency
- VQ2 = responsiveness
- VQ3 = downstream decision usefulness

Supervisor feedback was particularly important in identifying weaknesses in the research explanation. Questions about why consensus ranking was necessary, what was new in RDRA, how the five dimensions were justified, whether empirical reliability could be confused with stakeholder preference, and whether stability alone constituted ranking quality produced direct methodological improvements.

Another important lesson concerns research integrity and contribution boundaries. The current VQ3 results do not demonstrate an additional Top-3 advantage over Equal Borda, while the selected $\gamma=3.0$ refinement should not be presented as globally optimal or as replacing the primary RDRA methodology. Retaining these limitations makes the research contribution more credible and falsifiable.

At the end of Check Point 3, the primary methodological framework is substantially established, the major validation questions have been investigated, statistically supported findings have been distinguished from descriptive findings, unsupported generalisations have been removed, and complementary cross-task evidence has been obtained.

Therefore, the project has progressed from method development into research consolidation, contribution positioning and publication-oriented communication, with the remaining work focused on strengthening and communicating the established contribution rather than changing to a new research direction.

References

- [1] A. H. Alaoui, A. Meddaoui, and M. Hain, "AI-driven predictive maintenance for Industry 4.0: A systematic review of models, methods, and challenges," *The International Journal of Advanced Manufacturing Technology*, vol. 143, pp. 4861-4876, 2026, doi: 10.1007/s00170-026-17531-w.
- [2] T. Zonta, C. A. da Costa, R. da Rosa Righi, M. J. de Lima, E. S. da Trindade, and G. P. Li, "Predictive maintenance in the Industry 4.0: A systematic literature review," *Computers & Industrial Engineering*, vol. 150, Art. no. 106889, 2020, doi: 10.1016/j.cie.2020.106889.
- [3] A. Theissler, J. Pérez-Velázquez, M. Kettelgerdes, and G. Elger, "Predictive maintenance enabled by machine learning: Use cases and challenges in the automotive industry," *Reliability Engineering & System Safety*, vol. 215, Art. no. 107864, 2021, doi: 10.1016/j.ress.2021.107864.
- [4] O. Serradilla, E. Zugasti, J. Rodriguez, and U. Zurutuza, "Deep learning models for predictive maintenance: A survey, comparison, challenges and prospects," *Applied Intelligence*, vol. 52, no. 10, pp. 10934-10964, 2022, doi: 10.1007/s10489-021-03004-y.
- [5] H. Engbers and M. Freitag, "Automated algorithm selection for predictive maintenance: Advances and challenges," *Journal of Manufacturing Systems*, vol. 82, pp. 964-975, 2025, doi: 10.1016/j.jmsy.2025.06.023.
- [6] B. Taşçı, A. Omar, and S. Ayvaz, "Remaining useful lifetime prediction for predictive maintenance in manufacturing," *Computers & Industrial Engineering*, vol. 184, Art. no. 109566, 2023, doi: 10.1016/j.cie.2023.109566.

- [7] X. Bouthillier et al., "Accounting for variance in machine learning benchmarks," *Proceedings of Machine Learning and Systems*, vol. 3, pp. 747-769, 2021.
- [8] S. Matzka, "AI4I 2020 Predictive Maintenance Dataset," *UCI Machine Learning Repository*, 2020, doi: 10.24432/C5HS5C.
- [9] H. Wang, W. Luo, Z. Liu, J. Zhang, and M. Zuo, "Fourier feature refiner network with soft thresholding for machinery fault diagnosis under highly noisy conditions," *IEEE Internet of Things Journal*, vol. 11, no. 13, pp. 22880-22891, 2024, doi: 10.1109/JIOT.2024.3363216.
- [10] T. Silva Filho, H. Song, M. Perello-Nieto, R. Santos-Rodriguez, M. Kull, and P. Flach, "Classifier calibration: A survey on how to assess and improve predicted class probabilities," *Machine Learning*, vol. 112, pp. 3211-3260, 2023, doi: 10.1007/s10994-023-06336-7.
- [11] Y. Lv, X. Guo, Q. Zhou, L. Qian, and J. Liu, "Predictive maintenance decision-making for variable faults with non-equivalent costs of fault severities," *Advanced Engineering Informatics*, vol. 56, Art. no. 102011, 2023, doi: 10.1016/j.aei.2023.102011.
- [12] H. Liao, Y. He, X. Wu, Z. Wu, and R. Bausys, "Reimagining multi-criterion decision making by data-driven methods based on machine learning: A literature review," *Information Fusion*, vol. 100, Art. no. 101970, 2023, doi: 10.1016/j.inffus.2023.101970.
- [13] D. Boix-Cots, F. Pardo-Bosch, and P. Pujadas, "A systematic review on multi-criteria group decision-making methods based on weights: Analysis and classification scheme," *Information Fusion*, vol. 96, pp. 16-36, 2023, doi: 10.1016/j.inffus.2023.03.004.
- [14] A. Tornede, L. Gehring, T. Tornede, M. Wever, and E. Hüllermeier, "Algorithm selection on a meta level," *Machine Learning*, vol. 112, no. 4, pp. 1253-1286, 2023, doi: 10.1007/s10994-022-06161-4.
- [15] M. Mohammadi and J. Rezaei, "Ensemble ranking: Aggregation of rankings produced by different multi-criteria decision-making methods," *Omega*, vol. 96, Art. no. 102254, 2020, doi: 10.1016/j.omega.2020.102254.
- [16] H. Tyrallis and G. Papacharalampous, "A review of predictive uncertainty estimation with machine learning," *Artificial Intelligence Review*, vol. 57, Art. no. 94, 2024, doi: 10.1007/s10462-023-10698-8.
- [17] M. G. Kendall, "A new measure of rank correlation," *Biometrika*, vol. 30, nos. 1-2, pp. 81-93, 1938.
- [18] R. Kolde, S. Laur, P. Adler, and J. Vilo, "Robust rank aggregation for gene list integration and meta-analysis," *Bioinformatics*, vol. 28, no. 4, pp. 573-580, 2012, doi: 10.1093/bioinformatics/btr709.

- [19] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65-70, 1979.
- [20] A. Saxena and K. Goebel, "Turbofan Engine Degradation Simulation Data Set," *NASA Prognostics Data Repository*, NASA Ames Research Center, Moffett Field, CA, USA, 2008.
- [21] A. Saxena, K. Goebel, D. Simon, and N. Eklund, "Damage propagation modeling for aircraft engine run-to-failure simulation," in *2008 International Conference on Prognostics and Health Management*, Denver, CO, USA, 2008, pp. 1-9, doi: 10.1109/PHM.2008.4711414.